

ASSIGNMENT-2

$$A = 3, 4, 2, 5$$

$$B = 6, 7, 5, 8$$

$$C = 4, 3, 5, 4$$

$$E = 0.2$$

$$P(e) = 0.2$$

$$P(e) = 0.8$$

$$\text{Initially } N(A) = N(B) = N(C) = 0$$

$$Q(A) = Q(B) = Q(C) = 0$$

$$R = 3$$

$$N(A) = 1$$

Play 1

$$Q(A) = 0 + \frac{3-0}{1} = 3$$

$$Q(A) = 3 \quad Q(B) = 0 \quad Q(C) = 0$$

Play 2

$$Q(A) = 3 > Q(B) = 0 \quad Q(C) = 0$$

$$R = 4$$

$$N(A) = 2$$

$$Q(A) = 3 + \frac{4-3}{2}$$

$$3 + \frac{1}{2}$$

$$Q(A) = 3.5$$

play 3-

$$Q(A) = 3.5$$

$$R = 2$$

$$N(A) = 3$$

$$Q(A) = 3 + 5 + \frac{2 \cdot 3 \cdot 5}{3}$$

$$3.5 + \frac{1.5}{3}$$

$$\boxed{Q(A) = 3}$$

play 4

$$R = 4$$

$$N(A) = 4$$

$$Q(A) = 3 + \frac{5-3}{4}$$

$$Q(A) = 3.5$$

$$Q(A) = 3.5$$

$$Q(B) = 0$$

$$Q(C) = 0$$

$$P-1 = A$$

$$Q(A) = 0 + \frac{3-0}{1} = 3$$

$$P-2 = A \quad Q(A) = 3 + \frac{2-3}{2} = 3.5$$

$$P-3 = B \quad Q(B) = 0 + \frac{6-0}{1} = 6$$

$$P-4 = B \quad Q(B) = 6 + \frac{7-6}{2} = 6.5$$

$$P-5 = A \quad Q(A) = 3.5 + \frac{2 \cdot 3 \cdot 5}{3}$$

$$Q(A) = 3$$

$$P-6 = C \quad Q(C) = 0 + \frac{4-0}{1} = 4$$

$$P-7 = B \quad Q(B) = \frac{6.5 + 5 - 6.5}{1} = 6$$

$$Q(B) = 6$$

$$P-8 = A \quad Q(A) = 3 + \frac{5-3}{4} = 3.5$$

$$P-9 = C \quad Q(C) = 4 + \frac{3-4}{2} = 3.5$$

$$P-10 = B = Q(B) = 6 + \frac{8-6}{4} = 6.5$$

$$Q(B) = 6.5$$

N. Table $N(A) = 4$

$$N(B) = 4$$

$$N(C) = 2$$

$$R(A) = [3, 4, 2, 5]$$

$$R(B) = [6, 7, 5, 8]$$

$$R(C) = [4, 3]$$

Q Table $Q(A) = 3.5$

$$Q(B) = 6.5$$

$$Q(C) = ~~3~~ 3.5$$

$$\max(3.5, 6.5, 3.5)$$

Selected machine, $= 6.5$